



3.6.6 Wrap-Up: Error Analysis

1. Two students each evaluated an expression. Their work is shown in the table. Each student made a mistake. Circle the step where each student made an error and describe the mistake they made.

Problem	Describe the Mistake
$\left(\frac{1}{5}\right)^4 \cdot \left(\frac{1}{5}\right)^{-2} \cdot 5^{-1}$ $= \left(\frac{1}{5}\right)^{-8} \cdot 5^{-1}$ $= 5^8 \cdot 5^{-1}$ $= 5^7$ $= 78,125$	
$\left(\frac{3}{2}\right)^{-3} \cdot (2)^3$ $= \frac{2^{-3}}{3^{-3}} \cdot 2^3$ $= \frac{2^0}{3^{-3}}$ $= \frac{1}{3^{-3}}$ $= 3^3$ $= 27$	

Unit 3, Lesson 7: Equivalent Expressions



Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.



Learning Target

- ☐ I can determine equivalency using the laws of exponents.



3.7.1 Warm-Up: Bell Work

1. Evaluate each expression. Show your work.

a. $\frac{(-4)^2(-4)^{-3}}{-4}$

b. $\left(\left(\frac{2}{5}\right)^3\right)^{-2} \cdot \left(\frac{5}{2}\right)^{-4}$



3.7.2 Exploration: Exponential Expressions With Different Bases

1. Three expressions are given in the table.

- a. For each given expression, generate an equivalent expression by applying exponent laws. Then, find the value of the expression using a calculator.

Given Expression	Equivalent Expression	Value
$\frac{2^{14}}{2^4}$		
$4^7 \cdot 4^{-2}$		
$16 \cdot 4 \cdot 4^2$		

- b. Discuss your results with your partner.
What do you notice about the three expressions?



3.7.3 Guided Instruction: Rewriting Expressions With Different Bases

Exponent laws can be applied to exponents with the **same base**. However, sometimes a base can be rewritten to create a common base when perfect squares, cubes, fourths, etc. are present.

1. Complete the table. The first row is started for you.

Base	Square	Cube	Fourth
2	$2^2 = 4$	$2^3 = 8$	$2^4 = \underline{\hspace{2cm}}$
3			
4			
5			

2. Write an exponential expression equivalent to 8^5 that has a base of 2.

Step	Description
8^5	Determine an expression with base 2 that is equivalent to 8.
$= (\underline{\hspace{1cm}})^5$	Substitute the power of 2 for 8 in the expression.
$= 2^{\boxed{\hspace{1cm}}}$	Apply power of a power law of exponents.

3. Write an exponential expression equivalent to 9^4 that has a base of 3.

$$9^4$$

$$= (\underline{\hspace{1cm}})^4$$

$$= 3^{\boxed{\hspace{1cm}}}$$

4. Complete the steps to write an exponential expression equivalent to $2^3 \cdot 4^3$ that has a base of 2.

$$2^3 \cdot 4^3$$

$$= 2^3 \cdot (\boxed{\hspace{1cm}})^3$$

$$= 2^3 \cdot 2^{\boxed{\hspace{1cm}}}$$

$$= 2^{\boxed{\hspace{1cm}}}$$

5. Work with your partner to rewrite each expression from the Exploration using a common base of 2.

Expression A	Expression B	Expression C
$\frac{2^{14}}{2^4}$	$4^7 \cdot 4^{-2}$	$16 \cdot 4 \cdot 4^2$



3.7.4 Guided Practice: Different Bases, Same Value

The laws of exponents can be applied to determine if expressions are equivalent.

1. Consider the expressions $3^2 \cdot 6^3$ and $\frac{(3 \cdot 2)^5}{2^2}$.

- a. Complete the steps to rewrite each expression using laws of exponents.

Expression A	Expression B
$3^2 \cdot 6^3$	$\frac{(3 \cdot 2)^5}{2^2}$
$= 3^2 \cdot (\square \cdot 2)^3$	$= \frac{3^{\square} \cdot 2^{\square}}{2^2}$
$= 3^2 \cdot \square^{\square} \cdot \square^{\square}$	$= 3^{\square} \cdot 2^{\square}$
$= 3^{\square} \cdot 2^{\square}$	

- b. Complete the statement.

The expressions $3^2 \cdot 6^3$ and $\frac{(3 \cdot 2)^5}{2^2}$ are

☐ equivalent
☐ not equivalent

because they simplify to

☐ the same expression.
☐ different expressions.

2. Xavier says the expressions $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent because both expressions can be rewritten as 2^{-7} .

- a. Generate an equivalent expression for each expression using a base of 2. Show your work.

$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$	$\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$

- b. Explain whether or not you agree with Xavier.



3.7.5 Collaboration: Equivalent Expressions

1. With your partner, determine who is Partner A and who is Partner B.
 - a. Generate an equivalent exponential expression for each expression in your column using as few bases as possible.

Partner A		Partner B
$3^3 \cdot 3^4$		$\frac{3^9}{3^2}$
$\frac{2^{-2}}{2^{-4}} \cdot 7$		$\frac{(7 \cdot 2)^2}{7}$
$2^2 \cdot 4^2$		$4^5 \cdot 4^{-2}$
$\frac{3^5}{6^2}$		$3 \cdot \left(\frac{2}{3}\right)^{-2}$
$\frac{5^2 \cdot 5^0}{2^{-4} \cdot 2^3}$		$\frac{10^4}{5^2 \cdot 2^3}$

- b. The expressions in each row are equivalent. If your expression and your partner's expressions are not equivalent, work together to find and correct any error(s). Then, write the equivalent expression in the middle column.



3.7.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.



Unit 3, Lesson 8: Writing Large Numbers in Scientific Notation



Benchmarks

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.



Learning Targets

- ☐ I can use the laws of exponents to evaluate numerical expressions with a base of 10 and exponents greater than or equal to 0.
- ☐ I can express large numbers using scientific notation.
- ☐ I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.